Theorem
Let G ⊆ SX be a Frobenius group by the permutation definition (transitive permutations on X with at most one fixed point)
Then

[G:H] = |X| since there is a 1-1 relation

fixed points H^g <=> cosets gH

with cardinality |X| and [G:H] respectively

Hence

|K| = [G:H]
= |X|

since
|K| = |G / ∪ H^g|
= |G| - ∑ |H^g/{1}| sum over distinct H^g
= [G:H]|H| - |X|(|H| - 1)
= [G:H]

Observation

H subgroup of permutations with a **single fixed point** h(0) = 0

H^k H with just the items **permuted**

K ≃ ?? all permutations

G/K

Claim: The Frobenius kernel K is a closed under composition.

k· k' ∈ K for any k, k' ∈ K

Lemma:
The following are equivalent

a) K is closed under composition

b) **coset**: cosets are **unique**

kH ≠ k'H for each k, k' ∈ K

Hence G|H is precisely all the cosets of K

G|H = { kH | k ∈ K }

since |G|H| = |K| = [G:K]

c) symmetry: permutation πk: G|H → G|H permute uniquely
i.e. for any k, k' ∈ K

πk(xH) ≠ πk'(xH) for all k, k' ∈ H note: since πk(x) = πk'(x)
<=> k'k ∈ H^x
<=> k' = k^-1

d) coset group: each coset group H * aH contains an element of K
i.e.

k ∈ H * aH for some k ∈ K

==========
proof of Lemma

a <=> b
Note that

k', k ∈ aH
<=> kH = k'H
<=> kk'^-1 ∈ H

Hence a => b:

K is closed

=> if kH = k'H, then kk'^-1 ∈ H ∩ K = {1}
so k = k'

=> each kH is unique

Conversely, b => a:

each kH is unique

Suppose

kk' ∈ H^a for some a

=> k^a k'^a ∈ H

=> k^a H = (k'^-1)^a H
NOTE: k^a, k'^a ∈ K

=> k = (k'^-1) by uniqueness
i.e. k' = k^-1

Hence the only time kk' ∈ H^a is if they are inverses.

QED
==========

Properties

elements of K

If k ∈ K

k^g ∈ K **conjugate**

C(k) ⊆ K **center**
(k) ⊆ K

==========

proof

If K is closed:

every coset is of the form kH $k \in K$

every conjugate is of the form H^k $k \in K$

for each $h \in H$, there exists exactly one $a \in G$ s.t.

$k(aH) = h(aH)$

If K is not closed:

$kh \in K$ for some $k \in K$

$hk, kh, k^h \in K$ for some $h \in H$

there exists one $a \in H$ s.t.

$aH \cap K = \varnothing$

Main observation
For any $g, g' \in G$

$\pi g(a) = b$ <=> $gg' \in H^a$
$\pi g'(b) = a$ $g'g \in H^b$

<=> $g(h^b)g' \in H^a$ for any h
$g'(h^a)g \in H^b$

So want to prove that

$\pi k(a) \neq \pi k'(a)$ for all $k, k' \in K$

Observation: If $k'k \in H^a$, then

$\pi k(a) = \pi k'(a) = bH$

for some b. So

$k' h^b k \in H^a$
$k^{-1}(h^b) k \in H^a$ for all h

$H^a = \{ k'h^bk \mid h \in H \}$
$= \{ k^{-1}'h^bk \mid h \in H \}$

=> $k'hk = k^{-1}h'k$ for some h, h'
=> $k' \in k^{-1}H$

Equivalently

there exists k'k ∈ H^a <=> there exists k'k ∈ H

Since

k' k = h^a for some h, a

=> (a^-1 k' a)(a^-1 k a) = h
∈ K ∈ K

Observation: If kh ∈ K for some h ∈ H, then we have some suspicious relations
All of

k ∈ K
kh ∈ K
hk ∈ K since hk = h(kh)h^-1 ∈ K
hkh^-1 ∈ K

but

khk^-1 ∈ H^k

Note: if

kh(a) = a => h(a) = k^-1(a)
=> h^-1(b) = k(b) where b = h(a)
hk(b) = b



Observation: cosets kH
If K is a subgroup, then it must be that cosets G|H are **exactly** all the cosets kH
i.e.

unique kH ≠ k'H if k ≠ k' since kH = k'H => kk'^-1 ∈ H

all G|H = { kH | k ∈ K }

Hence

kh ∈ H^a for some a

i.e. for each h ∈ H except h = 1, there is a unique (a, b)

πk(a) = b πh(b) = a note: otherwise kh ∈ K

How to prove??

NOTE: If for some x

y = πk(x) $\in$ H * xH

=> there exists h $\in$ H such that πh(y) = x
=> πkh(x) = x
i.e. kh $\in$ H^x

Hence, there must be **AT MOST |H|** values of x such that

πk(x) $\in$ H * xH

since otherwise by pigeon hole there is at least one k such that

πk(x) $\in$ H * xH πk(x') $\in$ H * x'H

=> kh $\in$ H^x $\cap$ H^x'
contradiction

Q: why?

Observation: **Center C(k)** is a subgroup in K. Dual conjugation ^g: K $\rightarrow$ K
The center C(k) is a **subgroup** in K
Note that C(k) is non-trivial:

(k) $\subseteq$ C(k)

so |C(k)| > 1.

claim: C(k) $\subseteq$ K
proof: Suppose on the contrary that k^g = k for some g $\in$ H^a i.e. g(aH) = aH
Then

k(aH) = k(g(aH))
= g(k(aH)) since kg = gk

=> g has two fixed points aH and kaH which is a contradiction



QED claim

claim: C(k) has at least |H| elements.
Since if |C(k)| < |H| then

|k^G| = [G: C(k)] > [G:H] = |K|

which is a contradiction since k^G ⊆ K.

Any center C(g) is a subgroup.

Hence

C(k) is a subgroup ⊆ K

C(k) = ker ^k where ^k: K → K

|C(k)| >= |H|

Consider the **conjugation map ^g: K → K**
Then

ker ^k = C(k) for any k ∈ K

ker ^g = ∅ for any g ∉ K
i.e. g ∈ H^a for some a

i.e.

^k k ∈ K has **at least one** fixed point

^g g ∈ H^a for any a has **NO** fixed point

Observation: conjugation k^g
If k ∈ K

k^g ∈ K for any g ∈ G

Hence

k ∈ aH => hkh^-1 ∈ haH
i.e. every coset in H * aH has an element of K

Approach 2: view G as permutations
Then

Each h ∈ H has **one fixed point s** and is of the form

h = (s)(...n cycle ...)(...n cycle ...) .... N many where either
a) H is cyclic and N >= 0
b) H is not cyclic and N >= n

since H is a group of

permutations with **no fixed point**

if we ignore s.

Each k ∈ K has **no fixed points** and is of the form

k = (...n cycle ...)(...n cycle ...) .... N many since if k has a smaller cycle m < n

then k^m has m fixed points.

If x ∈ H^a and y ∈ H^b then

xy(b) = x(b) since y(b) = b

(xy)^-1(a) = y^-1(x^-1(a)) since x(a) = a
= y^-1(a)

and

xy(c) = c if x^-1(c) = y(c)
i.e. xy ∈ H^c

==========

Theorem: subcases

H even
Let G be a Frobenius group with **H even** i.e. |H| is even
Then K is closed under composition.

|H| = |X|
Let G be a Frobenius group acting on a set X with |X| <= |H| + 1.
Then in fact, it must be that |X| = |H| + 1 K is closed under composition.
WRONG?

|k| = |K|
For any k ∈ K, (k) ⊆ K so if |k| = |K| then K = (k)

==========

Proof
*) subcase: |H| = |X|
Let i0 be the fixed point of H.
Let i ≠ i0. Then

h(i) ≠ h'(i) for any h, h' ∈ H

=> |X| >= |H| + 1 +1 for i0
=> |X| = |H| + 1

Furthermore H has no fixed point on X/{i0}
Hence, for any i,j ∈ X/{i0}, there is a unique h ∈ H

h(i) = j

Let k ∈ K

*) subcase: Even
Suppose H fixes 0 ∈ X
Since H is even, H has an element σ **of order 2**
i.e. σ^2 = 1

The main idea is that this simplifies the problem significantly.
For each x ∈ X, let

σx = σ^g for some g(0) = x

Note that σx preserves the properties of σ

σx(x) = x fixed point x

|σx| = 2 since σx^2 = (gσg^-1)(gσg^-1) = 1

claim 1: if x ≠ y, then σxσy ∈ K

Suppose on the contrary that σxσy has a fixed point

σxσy(u) = u

=> v = σx(u) = σy(u)
=> σxσy(v) = v
i.e. v is also a fixed point

=> either σxσy = 1 or u = v
But x ≠ y so it must be u = v

=> u = σx(u) = σy(u)
i.e. u is a fixed point of σx and σy too, which is a contradiction

QED claim

Let

K' = { σxσ0 | x ∈ X }

Then

|K'| = |X| = |K|

=> K' = K

In particular, every element of K has the form σx σ0.

claim 2: K is closed under composition
Let k, k' ∈ K and

k = σxσ0 k'^-1 = σyσ0

=> k k'^-1 = σxσ0 (σyσ0)^-1
= σxσy ∈ K by claim

QED
==========

Transfer map
https://en.wikipedia.org/wiki/Transfer_(group_theory)
Let ai be representatives of H cosets.
Then for any g ∈ G and ai,

g · aiH = ai'H
g · ai = ai' · h_g,i for some h_g,i ∈ H

The **transfer map** is

V: G → H/H' where H' = [H,H]
V(g) = ∏ h_g,i mod H'

This is **independent** of the choice of representatives.
i.e. If bi = ai · hi is another set of representatives, then

g · bi = g · (ai · hi)
= (ai' · h_g,i) · hi by definition of g · ai
= bi' · (hi'^-1 · h_g,i · hi)

=> ∏ (hi'^-1 · h_g,i · hi)
= ∏ h_gi by cancellation mod H'
= V(g)

If H is **malnormal**, then

V(h) = hH'

==========
proof

- V(h) = hH' if H malnormal
Let h ∈ H and

h · ai = ai' · h_h,i

=> h_h,i = ai'^-1 · h · ai note: i ≠ i' for all ai ∉ H since H^ai ∩ H = { 1 }

Hence

V(h) = ∏ h_h,i

= ∏ ai'^-1 · h · ai

= (h'^-1 · h · h') (a^-1 · h^[G:H]-1 · a) Most ai **cancel** each other

(ai'^-1 h ai)(ai^-1 h ai'')
= ai^-1 h^2 ai''

EXCEPT

1) **last ai = a** remaining after cancelation

2) when i = i'
i.e. when ai = h' ∈ H

= (h'^-1 · h · h') · (a^-1 · h^[G:H]-1 · a)
= h (mod H') **= 1 (mod H')**
since H ∩ H^a = { 1 }

= h (mod H')

=> V(h) = h mod [H,H]

QED
==========

==========

== Fixed point ==

Irreducible permutations
Let G ⊆ SX be a group of permutations of X.
G is **reducible** if there is a X' ⊆ X such that

G|X' ≃ G G|X' = G restricted to X'

G is **irreducible** otherwise.

e.g. if

h = (0)(1 2 3) H = (h) is irreducible

h' = (0)(1 2 3)(4 5 6) H' = (h') is reducible
since H' ≃ H and H = H'|Z4

No fixed point permutation
What does a permutation group with no fixed point look like?

Let H ⊆ Sn be a group with no fixed point
Then each for an i ≠ 0, the

h(i) ≠ h'(i) since if h(i) = h'(i), then h^-1h' has a fixed point at i

Theorem: cycles
Let H be a permutation group with no fixed points.

cyclic decomposition

Each h ∈ H must be decomposable to **disjoint cycles** of exactly the **same size**
i.e.

h = **(.... n cycle ...) (.... n cycle ...) ...**

Furthermore, either:
1) H is **cyclic**
A generator h is of the form

h = **(... n cycle...) (... n cycle 2...) ....**

H = (h)

If we consider only irreducible permutations

H is an **irreducible n cycle** <=> H = (h)
h = **(... n cycle ...)**

2) H is not cyclic
Each h is composed of **N >= n** many **n-cycles**

h = (... n cycle 1...) (... n cycle 2...) .... (... n cycle N ...) N >= n

However, not all h may be composed of cycles of the same length (example below)

Uniqueness

If H ◁ Sn is irreducible then for any pair 0 <= i, j <= n-1 not equal

h(i) = j **unique** h ∈ H
for any pair 0 <= i, j <= n-1 i ≠ j

Let

hi ∈ H where hi(0) = i

Hence

H = {1,h1,... hn-1}

|H| = n

Representation
Any h can be uniquely written as

h = hi0 ◦ h0^-1 i0 where h(0) = hi0(0)

= hi1 ◦ h1^-1 i1 where h(1) = hi1(1)

= hi2 ◦ h2^-1 i2 where h(2) = hi2(2)
...

==========
proof
a) cycles of the same size.

h = **(.... n1 cycle ...) (.... n2 cycle ...) ...** n1 < n2

=> h(x) = h^n1(x) for any x ∈ n1 cycle
but h^n1 ≠ 1

2) is because: suppose H is not a cycle and N < n, and h and h' are elements, not in the same cycle.
Then by the pigeon hole principle, there is at least one cycle containing 2 same elements

h = (... x ...y ..) ....

h' = (x .... y ...) ....

since each cycle is length n but there are only N < n cycles.

=> for some i, j y = h^i(x) = h^j(x)

which is impossible (since h(x) must be distinct for each h $\in$ H)

QED
==========

Example: h composed of 3 cycles only
Note: hi: Z9 → Z9 defined by

hi(0) = i

Let

h0 = 1

h1 = (0 1 2)(3 4 5)(6 7 8)
h2 = h1^2
= h1^-1
= (0 2 1)(3 5 4)(6 8 7)

h3 = (0 3 6)(1 4 7)(2 5 8)
h6 = h3^2
= h3^-1
= (0 6 3)(1 7 4)(2 8 5)

h4 = h3 h1
= (0 4 8)(1 5 6)(2 3 7)
h8 = h4^2
= h4^-1
= (0 8 4)(1 6 5)(2 7 3)

h5 = h4 h1
= (0 5 7)(1 3 8)(2 4 6)
h7 = h5^2
= h5^-1
= (0 7 5)(1 8 3)(2 6 4)

All elements have the same order:

|hi| = 3 for all i

|H| = 8

Unique **decompositions**:

h1	= h1 1^-1	= h1	0 → 1
	= h2 h1^-1	= h1	1 → 2
	= 1 h2^-1	= h1	2 → 0
	= h4 h3^-1	= h4 h6	3 → 4
	= h5 h4^-1	= h5 h8	4 → 5
	= h3 h5^-1	= h3 h7	5 → 3
	= h7 h6^-1	= h7 h3	6 → 7
	= h8 h7^-1	= h8 h5	7 → 8
	= h6 h8^-1	= h6 h4	8 → 6

h3	= h3 1	= h3^-1	0 → 3
	= h4 h1^-1	= h4 h2	1 → 4
	= h5 h2^-1	= h5 h1	2 → 5
	= h6 h3^-1	= h3	3 → 6
	= h7 h4^-1	= h7 h8	4 → 7
	= h8 h5^-1	= h8 h7	5 → 8
	= 1 h6^-1	= h3	6 → 0
	= h1 h7^-1	= h1 h5	7 → 1
	= h2 h8^-1	= h2 h4	8 → 2

Example: H composed of cycles of different lengths
i.e. H has elements of different **order**
note: hi: Z6 → Z6 is defined by

hi(0) = i

h0 = 1

h1 = (0 1)(2 3)(4 5)

h2 = (0 2)(1 4)(3 5)

h4 = h2 h1
= (0 4 3)(1 2 5)
h3 = h4^2
= h1 h2
= (0 3 4)(1 5 2)

h5 = h2 h4
= (0 5)(1 3)(4 2)

(0 1)(2 3)(4 5)

	0	1	2	3	4	5
0		1				
1	1					
2				1		
3			1			
4						1
5					1	

(0 2)(1 4)(3 5)

	0	1	2	3	4	5
0			1			
1					1	
2	1					
3						1
4		1				
5				1		

(0 4 3)(1 2 5)

	0	1	2	3	4	5
0				1		
1						1
2		1				
3					1	
4	1					
5			1			

(0 3 4)(1 5 2)

	0	1	2	3	4	5
0					1	
1			1			
2						1
3	1					
4				1		
5		1				

(0 5)(1 3)(4 2)

	0	1	2	3	4	5
0						1
1				1		
2					1	
3		1				
4			1			
5	1					

Hence different orders

|h1| = |h2| = |h5| = 2

|h3| = |h4| = 3

|H| = 6

Unique **decompositions**:

h1	= h1 1	= h1	0 → 1
	= 1 h1^-1	= h1	1 → 0
	= h3 h2^-1	= h3 h2	2 → 3
	= h2 h3^-1	= h2 h4	3 → 2
	= h5 h4^-1	= h5 h3	4 → 5
	= h4 h5^-1	= h4 h5	5 → 4
h5	= h5 1	= h5	0 → 5
	= h3 h1^-1	= h3 h1	1 → 3
	= h4 h2^-1	= h4 h2	2 → 4
	= h1 h3^-1	= h1 h4	3 → 1
	= h2 h4^-1	= h2 h3	4 → 2
	= h5^-1	= h5	5 → 0

==========

== Examples ==

Example

Generated by

h = (a b c d)
a = (h a b d c)

	h	a	b	c	d
h	h	b	c	d	a
h^2	h	c	d	a	b
h^3	h	d	a	b	c
a	a	b = h(a)	d = h^2(b)	h	c = h^3(d)

Note that

a*(a) = h(a)
a*(b) = h^2(b)
a*(d) = h^3(d)

Hence

	set	**h**	**a**	**b**	**c**	**d**
b = a^2	K	b	d =h^3	c =h	a =h^2	h
d = a^3	K	d	c =h^2	h	b =h^3	a =h
c = a^4 = a^-1	K	c	h	a =h^3	d =h	b =h^2
ah	H^c	a =a	d =h^3	h =d	**c**	b =h^2

ah^2	H^d	a =a	h =c	c =h	b =h^3	**d**
ah^3	H^b	a =a	c =h^2	**b**	d =h	h =b
bh	H^d	b =b	c =h^2	a =h^3	h =a	**d**
bh^2	H^a	b =b	**a**	h =d	d =h	c =h^3
bh^3	H^c	b =b	h =c	d =h^2	**c**	a =h
dh	H^b	d =d	h =c	**b**	a =h^2	c =h^3
dh^2	H^c	d =d	b =h	a =h^3	**c**	h =b
dh^3	H^a	d =d	**a**	c =h	h =a	b =h^2
ch	H^a	c =c	**a**	d =h^2	b =h^3	h =b
ch^2	H^b	c =c	d =h^3	**b**	h =a	a =h
ch^3	H^d	c =c	b =h	h =d	a =h^2	**d**

Example: C7 ⋊ C3

i.e.

K = C7

H = C3

H

h = (0)(1 2 4)(3 6 5) fixed point 0

h^2 = (0)(1 4 2)(3 5 6) fixed point 0

k = (0 1 2 3 4 5 6)

kh = (0 1 3)(2 5 4)(6) fixed point 6

kh^2 = (0 1 5)(2)(3 6 4) fixed point 2

k^2 = (0 2 4 6 1 3 5)

k^2h = (0 2 6)(1 4 3)(5) fixed point 3

k^2h^2 = (0 2 3)(1 6 5)(4) fixed point 4

k^3

... etc

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== Character theory ==

[Tao] Frobenius groups: https://terrytao.wordpress.com/tag/frobenius-groups/
[Tao] Frobenius group proof: https://terrytao.wordpress.com/2013/04/12/the-theorems-of-frobenius-and-suzuki-on-finite-groups/

$\bar{}(a+ib) = a-ib$

$E_{x \in G} f(x) = 1/|G| \sum_{x \in G} f(x)$ average

L2(G) is the Hilbert space of functions $G \to C$
i.e.

$f: G \to C \in L2(G)$

dot product $f \bullet g = E_{x \in G} f(x) \bar{g}(x)$

norm $||f|^2 = f \bullet f$

A basis of L2(G) is

$\xi g = \delta g$ i.e. $\delta g(x) = 1$ if x=g, 0 o.w.

hence

dim (L2(G)) = |G|

class functions L2(G)^G
L2(G)^G is the subspace of class functions $f \in L2(G)$ invariant under conjugation
i.e.

$f(x^g) = f(x)$ for any $g \in G$

$f|x^G = \{ f(g) \}$

Hence, a basis of L2(G)^G is

$\xi g1, ..., \xi gn$ where each gi^G is a unique conjugacy class

hence

dim L2(G)^G = # of conjugacy classes
i.e. **class number** of G

Unitary operator
$\rho \in U(V)$

$\rho(v) \bullet \rho(u) = v \bullet u$

If λ is an eigenvector, then

$\lambda v \bullet \lambda v = \rho(v) \bullet \rho(v)$
$= v \bullet v$

$\Rightarrow \lambda^2 = 1$

e.g.
If ρ = shift operator

$\rho(g)\ f(x) = f(g^{-1}x)$

$\Rightarrow \chi\rho(g) = \mathrm{tr}(\rho(g))$

$= \mathrm{tr}(\text{I shifted by k})$

$= \begin{cases} |G| & \text{if } g=1 \\ 0 & \text{o.e.} \end{cases}$

==========

```
This
is
code
Monaco 14
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